

# Water Resources Research®

## RESEARCH ARTICLE

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### Key Points:

- A model hybridizing deep learning and physics of dominant hydrological processes was developed for runoff simulation in alpine catchments
- Physics-informed deep learning model can well simulate runoff change under warming of 1°C, while the standard deep learning model can not
- Physics-informed deep learning outperforms physically-based models in simulating runoff response to climate change in alpine catchments

### Supporting Information:

Supporting Information may be found in the online version of this article.

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## Developing a Physics-Informed Deep Learning Model to Simulate Runoff Response to Climate Change in Alpine Catchments

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**Abstract** Climate change has rapidly degraded the cryosphere in alpine headwaters, altering the runoff regime of alpine watersheds and threatening water security worldwide. To precisely simulate runoff response to climate change in alpine catchments, we took the source region of the Yellow River as a case to develop a physics-informed deep learning (DL) model that tightly hybridizes DL with the physics of dominant hydrological processes, including the neural network-based coupling of soil freeze-thaw physics inspired by the Stefan equation. A physics-informed DL model without soil freeze-thaw representation was also established as a benchmark. Results demonstrated the former model's superiority in capturing streamflow's inter-annual dynamics and reproducing baseflow recession properties caused by soil freeze-thaw when average temperature rises by 1°C. This emphasizes the importance of the physics of dominant hydrological processes for DL to make credible runoff projections under rapid climate change. Furthermore, we comprehensively compared the physics-informed DL model with a lumped model, physically-based distributed model, and standard DL model, showing that the physics-informed DL model performs best in simulating streamflow at multi-timescales and the impacts of soil freeze-thaw on runoff, and is more capable of simulating runoff relative change and trend in response to climate change. In summary, our findings demonstrate physics-informed DL's credibility in runoff projections under drastic climate change when infused with the physics of dominant hydrological processes, and illustrate its superior ability to learn complex hydrological processes, providing a new approach for runoff simulation in permafrost-affected alpine catchments under climate change.

**Plain Language Summary** The changing climate greatly affects permafrost regions in mountain headwaters, leading to the shift of the runoff regime. Projecting future runoff under climate change in such catchments is crucial for global water security. However, traditional distributed hydrological modeling falls short due to the complex and unclear underlying mechanisms, while deep learning (DL) models' unreliability in out-of-sample predictions might be amplified by significant climate change. To address this gap, we focused on an alpine permafrost catchment, the source region of the Yellow River, and developed a model that combines DL techniques with the fundamental physics of hydrological processes. By introducing a neural network-based coupling of soil freeze-thaw, inspired by a well-known physical equation, our model outperforms the benchmark model without soil freeze-thaw representation in simulating annual runoff variations in response to a 1°C/year temperature increase. Furthermore, our model excels in simulating runoff changes and trends caused by climate change, surpassing the lumped model, distributed hydrological model, and standard DL model. These indicate that the infused physics can help DL better capture the soil freeze-thaw process and make more credible projections under climate change. Our study offers a valuable approach for assessing climate change impacts on runoff in permafrost-affected mountain headwaters.

## 1. Introduction

The alpine headwater is an important water supply in the world, with an ecosystem very vulnerable to climate change due to the high altitude, cold temperature, and slow vegetation growth (G. Wang et al., 2007). With the global temperature rising at a rate of 0.2°C per decade (IPCC, 2013), the cryosphere in alpine catchments, which is an amplifier of global warming (Hu et al., 2021), has been significantly degraded, with permafrost thawing, snow cover declining, and glaciers retreating (Beniston et al., 2018; Gao et al., 2021; Qin et al., 2017). This has altered surface and subsurface hydrological processes, further threatening the security of water resources.

Therefore, studying the runoff response to climate change is fundamental for water resource management in these and downstream regions.

Nevertheless, due to the complexity of hydrological processes and their interactions, the hydrological modeling in alpine catchments presents unique challenges, particularly given the rapidly changing climate. Frozen soil, snow, and glaciers are widely distributed in alpine catchments. They are the most critical processes in the research of cryosphere hydrology. Even though there have been more focused studies in recent years, there is still a knowledge gap regarding the hydrology regime of the processes at various scales and the interaction between the processes (Beniston et al., 2018; Gao et al., 2021; Rush & Rajaram, 2022). Taking permafrost hydrology studies as an example, the oxygen isotope observation in northern Switzerland suggested that permafrost has negligible impacts on runoff (Lindström et al., 2002). However, a large number of basins with permafrost, including northern Canada (Abdul Aziz & Burn, 2006), Northwest China (Liu et al., 2007), and Northeast China (Liu et al., 2003), have seen a significant increase in winter baseflow which is generally attributed to the enlargement of groundwater storage in the rainy season due to permafrost degradation. A possible explanation for these inconsistent findings is that, against the backdrop of climate warming, though the liquid water resulting from permafrost thawing can increase groundwater and therefore recharge runoff, this effect may be partially offset by increased runoff losses caused by increased soil evaporation and infiltration (G. Wang & Liu, 2012). So, the ultimate impact of permafrost degradation on runoff varies with the characteristics of different drainage basins. Moreover, the interaction between frozen soil, snow cover, and glaciers further complicates alpine hydrology and requires more research to be fully understood. For example, according to conventional understanding, snow cover insulates the soil from the cold winter air, inhibiting the growth of frozen soil. In contrast, a recent study found that the cold snowpack can act as a heat sink and promote frozen ground during the snowmelt season (Rush & Rajaram, 2022).

Currently, the hydrological simulation at the basin scale in alpine regions is frequently performed using physically-based models such as the CRHM (DeBeer & Pomeroy, 2017; J. Zhou et al., 2014), SWAT (Gu et al., 2022; K. Li, Wang, et al., 2021; Zeng et al., 2022), VIC (S. Li, Wang, et al., 2021; Y. Wang, Zhang, et al., 2020), Alpine3D (Brauchli et al., 2017), GBEHM (B. Gao, Qin, et al., 2016), CoupModel (J. Zhou et al., 2013), WaSiM (A. Sun et al., 2020), Topkapi-ETH (Fatichi et al., 2015), and HGS (Thornton et al., 2022), and so on. Most of these physically-based models are based on energy balance, water balance, and heat transfer equations to describe cryospheric processes. However, some challenges arise when these models attempt to represent the cryospheric processes as realistically as possible, one of which is equifinality (Beven, 2006). The two main sources of equifinality are generally considered to be the error in the model structure (Beven, 2006; McDonnell et al., 2007) and the uncertainty in the parameterization (Kuppel et al., 2018), which are more significant in alpine catchments. On the one hand, given the complexity of hydrological processes in the cryosphere, our imperfect understanding of these processes may result in biased model generalization and thus increase the model error, including but not limited to the neglect of redistribution, turbulent heat exchange in snow models (Dong, 2018; Griessinger et al., 2019), inappropriate representation of moisture profiles and hydraulic conductivity in air-water-ice three-phase mixed soils (H. Jiang, Zheng, Yi, et al., 2020; Walvoord & Kurylyk, 2016), solving heat transfer equations limited to the vertical direction when coupled with hydrological models (Z. Sun et al., 2019), and inapplicability of existing water routing algorithms in permafrost zones due to the uncertainty in river ice-jam frequency and intensity (Walvoord & Kurylyk, 2016). In addition, the huge heterogeneity in temperature, radiation, and landscape, as well as possible scale effects in processes reported by Gao et al. (2021) in alpine catchments are other sources of the model error. On the other hand, the information from streamflow alone is often not sufficient for the parameterization of these physically-based models. Additional data such as landscapes, soil properties, soil temperature, and soil moisture are required to constrain the parameters associated with different processes in the model (Kuppel et al., 2018) or to determine the initial state and boundary conditions (H. Jiang, Zheng, Yi, et al., 2020). Unfortunately, surface and borehole observations in alpine catchments are scarce due to the harsh environment and technical barriers. Despite the incipient potential of remote sensing data combined with data assimilation for model parameterization, significant uncertainties still exist, especially in alpine catchments (Che et al., 2019; H. Jiang, Zheng, Yi, et al., 2020). These deficiencies in data quality may convey inconsistent information to the model and pull the model in different directions during calibration (Kuppel et al., 2018), thus exacerbating equifinality.

In recent years, deep learning (DL), a method that differs from conventional physically-based models, has recently been introduced into the field of hydrology (Shen et al., 2018; Tao et al., 2016), showing great potential in runoff simulation and gradually achieving a more accurate simulation than physically-based models (Kratzert

et al., 2018; Kratzert, Klotz, et al., 2019). With little or no expert knowledge, DL is effective at learning the underlying principles from vast amounts of hydrological data and applying them to streamflow modeling. It is even argued that DL may be able to acquire the knowledge that hydrologists are unable to translate into theory or process-based models (Nearing et al., 2021). The credibility of DL under climate change, however, is disputed by hydrologists (Nearing et al., 2021), since DL makes inferences based on the assumption that the distribution of the dataset in the prediction period is consistent with that in the training period. This assumption may not be satisfied, especially in alpine catchments with a rapidly changing climate.

A feasible attempt to enhance DL's credibility in runoff projections under climate change is to incorporate our process knowledge into DL models, which is generally referred to as physics-based, physics-guided, physics-constrained, or physics-informed DL (He et al., 2020; Shen et al., 2021). The plausibility of the physics-informed DL is based on the consensus that the infused physics often represent hydrologists' understanding of underlying laws in geosystem dynamics, such as the conservation of mass and energy, which are considered to hold true even under a non-stationary climate and can therefore improve DL's consistency in out-of-sample prediction. Previous studies have also demonstrated that physics-informed DL models outperform purely data-driven models when applied to extrapolations outside of the training data (Shen & Lawson, 2021). The approaches to integrating physics with DL include imposing physical constraints into loss functions (Read et al., 2019; N. Wang, Zhang, et al., 2020; K. Xie et al., 2021), constructing a component that conserves mass and energy (Kraft et al., 2021; Zhao et al., 2019), feeding the output of physical hydrological models into neural networks to correct the mismatch (Cho & Kim, 2022; Konapala et al., 2020; S. Yang et al., 2020), and employing machine learning approaches to find parameter sets for physical models (Feng et al., 2022; Tsai et al., 2021). Recently, S. Jiang, Zheng, Yi, et al. (2020) achieved a closer symbiotic integration of the physics and DL, by implanting numerical equations of hydrological models into DL and training the physically meaningful parameters of hydrological models and the statistical parameters of DL models together. In this way, physical mechanisms guide AI to recognize fundamental hydrological theory, and AI can also compensate for knowledge gaps and data quality, thus promising to solve the hydrological simulation challenges under climate change. For the above hybrid framework, we argue that the infused physics needs to be given in accordance with the dominant hydrological processes in the study area, which may be sometimes challenging when the physical understanding of some dominant processes is insufficient.

From the developments above, this paper aims to develop a physics-informed DL model which considers the dominant hydrological processes in alpine catchments. Meanwhile, in order to evaluate the model's performance in the rapidly changing climate, the physics-informed DL model will be comprehensively compared with the lumped model, physically-based model, and standard DL model, particularly in terms of the model's capacity to simulate runoff response to climate change.

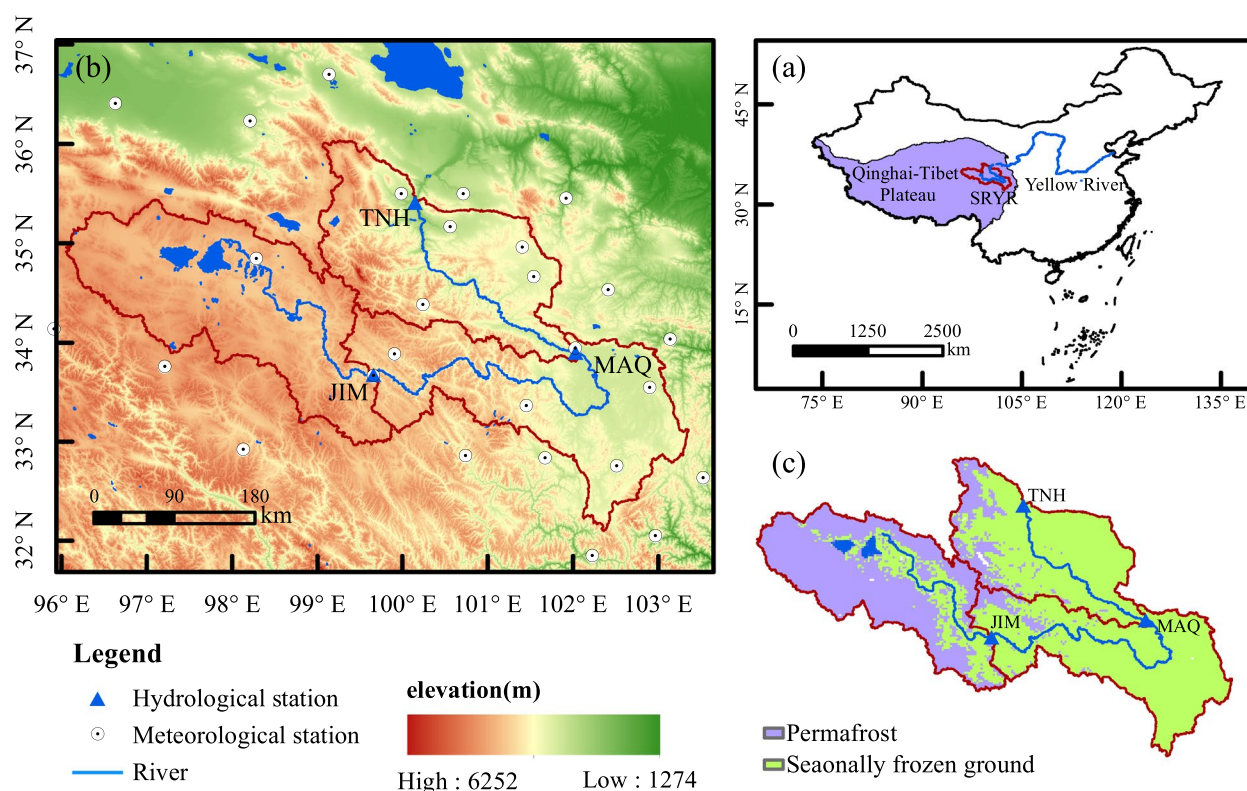
## 2. Materials and Methods

### 2.1. Study Area and Data

#### 2.1.1. Study Area

The source region of the Yellow River (SRYR), that is, the catchment of the Yellow River Basin above Tangnaihai (TNH) hydrological station, is located in the northeast of Qinghai-Tibetan Plateau (QTP) with an area of 124,000 km<sup>2</sup> (Figure 1). The SRYR, regarded as one of Asia's water towers, provides 38% of runoff but only covers 15% of the Yellow River basin. The elevation of the SRYR ranges from 2,656 m to 6,252 m, making it an amplifier of global climate change—the annual mean air temperature in this region has increased at the rate of 0.33°C per decade since 1960, which is almost twice as fast as the global temperature warming. The rapid temperature increase accelerates the degradation of the cryosphere in the SRYR, as evidenced by a decrease in the maximum thickness of seasonal permafrost of 0.012 m per year and a decline in the area ratio of permafrost of 1.1% per year during 1981–2015 (Qin et al., 2017), shorter snow accumulation periods and days of snow cover from 1978 to 2016 (X. Liu, Zhang, et al., 2020), and a glacier area shrank from 125 km<sup>2</sup> in 1960 to 104 km<sup>2</sup> in 2000 (J. Yang et al., 2003). Thus, the study on the runoff response under climate change in the SRYR is also of great reference significance for other alpine mountains.

The SRYR features a continental alpine climate in its three subbasins, with mean annual temperatures ranging from −4.0 to −2.0°C and mean annual precipitation ranging from 468.1 to 535 mm during the period of



**Figure 1.** (a) Relative location of the study area; (b) distribution of meteorological stations and hydrological stations in the study area; (c) frozen ground distribution, adapted from Zheng et al. (2020).

1960–2019. According to the locations of Jimai (JIM), Maqu (MAQ), and Tangnaihai (TNH) hydrological stations, the SRYR can be divided into three subbasins of different sizes, which are shown in Table 1. It should be noted that the basin above the JIM station is dominantly covered by permafrost, and the downstream basins are dominated by seasonally frozen ground. Previous studies have shown that runoff reduction at the JIM station is mainly due to the increase of surface runoff wastes (evaporation and infiltration) caused by the degradation of the permafrost, while in the region between the JIM and TNH, it is also affected by the changes in the position of the southeastern warm–wet current arriving (Lan et al., 2010; S. Zhang, 2004). Therefore, the runoff simulation at the JIM station is more typical for examining the impacts of permafrost degradation on runoff.

### 2.1.2. Data

Daily meteorological data were collected from 27 meteorological stations within and around the SRYR, including air temperature (*tas*), precipitation (*pr*), relative humidity (*rhu*), wind speed (*win*), sunshine duration (*ssd*), and air pressure (*ps*), which could be downloaded from the National Meteorological Information Center of China

**Table 1**  
*Geomorphology, Climate, and Permafrost Characteristics in Three Subbasins*

| Subbasin                                                                     | Above JIM   | Above MAQ   | Above TNH   |
|------------------------------------------------------------------------------|-------------|-------------|-------------|
| Area (km <sup>2</sup> )                                                      | 45,300      | 86,300      | 124,000     |
| Elevation (m)                                                                | 3,951–5,320 | 3,406–5,334 | 2,656–6,252 |
| Annual mean temperature (°C)                                                 | −4.0        | −2.4        | −2.0        |
| Annual mean precipitation (mm)                                               | 468.1       | 535.3       | 507.0       |
| The area ratio of permafrost (%)                                             | 76.9        | 43.5        | 38.2        |
| The change rate of maximum thickness of seasonally frozen ground (cm/decade) | −4.12       | −3.06       | −3.38       |

*Note.* The change rate of maximum thickness of seasonally frozen ground during 1965–2014 is from T. Wang et al. (2018).

Meteorological Administration (<http://data.cma.cn>). Potential evapotranspiration ( $pet$ ) was calculated using the FAO Penman-Monteith equation (Allen et al., 1998) based on the air temperature, relative humidity, wind speed, sunshine duration, and air pressure. Moreover, all these forcing data were spatially interpolated with an angular distance weighting method with elevation corrections (D. Yang et al., 2004) and then used to calculate basin-averaged values. Daily discharge data from 1960 to 2019 at the JIM, MAQ, and TNH stations were obtained from the Information Center of the Chinese Ministry of Water Resources, with the 1983 and 1990 data at the JIM station missing. The shuttle radar topography mission (SRTM) digital elevation model (DEM) with a resolution of 90 m (Jarvis, 2008) was used in this study. The distribution data of permafrost were from Zheng et al. (2020).

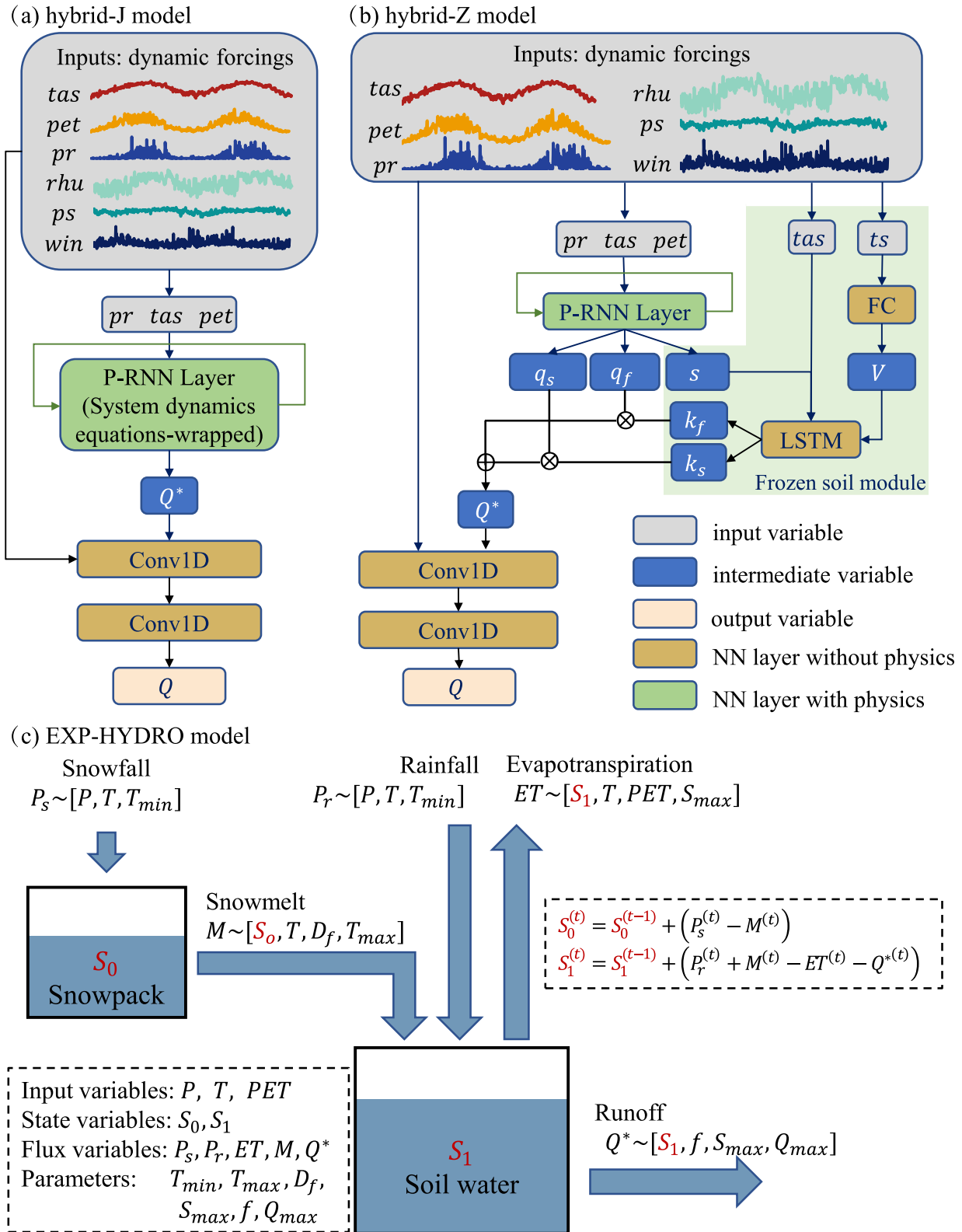
## 2.2. Models

### 2.2.1. Development of a Physics-Informed DL Model in Alpine Catchments

In this study, physics and DL were integrated using the hybrid framework developed by S. Jiang, Zheng, Yi, et al. (2020), which encodes physical geosystem dynamics as a physical process-wrapped recurrent neural network (denoted as the P-RNN layer), as shown in Figure 2a. In contrast to the recurrent neural network (RNN) (Rumelhart et al., 1986), in which neurons (inputs, states, and outputs) are connected by mathematical expressions, the connections in the P-RNN are specified in explicit discrete representations of the system dynamics (such as physical equations describing soil water movement, snow melting, etc.). Here, the dynamic geosystem incorporated into DL is represented by the exponential bucket hydrologic model (EXP-HYDRO), a conceptual, simple spatially lumped rainfall-runoff model developed by Patil and Stieglitz (2014). Specifically, we rewrote the EXP-HYDRO model using a differentiable framework in PyTorch (Paszke et al., 2019), and used the physical equations and parameters of the physical model to replace the mathematical expressions, as well as the weights and biases in the typical RNN architecture. Although not a process-based model, the EXP-HYDRO model (shown in Figure 2c) consists of two state buckets to represent snow water and soil water and has six physically meaningful parameters, making it simple and feasible to encode into the P-RNN layer. In this way, the P-RNN has a form similar to the ordinary RNN, except that it forgets, remembers, and outputs information from the sequences based on physical equations in the EXP-HYDRO model, making the output variable ( $Q^*$ ) also physically meaningful. After the P-RNN, a two-layer one-dimensional convolutional (Conv1D) network (Fukushima & Miyake, 1982), which is fed with the complete forcing and intermediate variable ( $Q^*$ ), is used to learn the processes not represented by the EXP-HYDRO model and correct the mismatch between the output of the lumped model and the observation streamflow ( $Q$ ). With this framework, the model hybridizes the physical approaches and data-driven components well and succeeds in combining the physical interpretability and consistency with the powerful learning capabilities of DL, which has been verified in 671 watersheds across the United States. For nomenclature's sake, the original model developed by S. Jiang, Zheng, Yi, et al. (2020) is later referred to “hybrid-J model”. Note that the inputs of the original EXP-HYDRO model are the precipitation, air temperature, and day length, where the day length is used to calculate the potential evapotranspiration using Hamon's formula (Hamon, 1963). In order to make full use of the physics in observed data, we calculated the Penman-Monteith evapotranspiration based on field data and used it as input to replace the Hamon evapotranspiration.

As seen, the physical knowledge infused into the hybrid-J model only includes snow accumulation and melting process but lacks the representation of frozen soil and glacier process since the EXP-HYDRO model is not specifically developed for the cryosphere. In the SRYR, there are only a few small glaciers that have a negligible impact on runoff (J. Wu, Liang, et al., 2020; J. Yang et al., 2003). While the soil freeze-thaw process is one of the most dominant hydrological processes and, therefore, must be taken into account in the development of the physics-informed DL model. Although the data-driven component following the P-RNN layer might have the ability to learn unrepresented processes, the Conv1D network is more often used to learn the short-term lagged influence of predictors on hydrological responses (Feng et al., 2020; S. Jiang, Zheng, Yi, et al., 2020), such as the flow routing process. In contrast, soil freeze-thaw is a long-term heat accumulation and consumption process and has extremely complex impacts on runoff, which may be beyond the capacity of the two-layer Conv1D network. Therefore, it is essential to incorporate a physical understanding of the soil freeze-thaw process into the physics-informed DL model to aid in its learning of these hydrological mechanisms.

In this case, the Stefan equation (Qin et al., 2018; Stefan, 1891), an empirical algorithm for calculating freezing depth based on the dynamic surface temperature, the volumetric water content of liquid water, and soil thermal conductivity (see Text S1 in Supporting Information S1), is used as a representation of the soil freeze-thaw



**Figure 2.** Model structure diagram. (a) The hybrid-J model. The P-RNN layer is encoded by the EXP-HYDRO model. (b) The hybrid-Z model. (c) The schematic diagram of the EXP-HYDRO model (adapted from S. Jiang, Zheng, Yi, et al., 2020).

process and infused into the physics-informed DL model. Inspired by the Stefan equation, a frozen soil module (Figure 2b) comprising an FC and LSTM layer (Hochreiter & Schmidhuber, 1997) is designed to use the inputs of the Stefan equation to characterize the freezing depth and learn the impacts of freezing depth on runoff. Specifically, we use the air temperature ( $t_{as}$ ) to represent the external energy exchange with frozen soil. In the Stefan equation, the conversion of air temperature to soil temperature is characterized by an  $n$ -factor, which is often empirically determined as a constant (Gao et al., 2022; T. Wang et al., 2018). However, in fact,  $n$ -factor presents obvious periodic changes under the influence of vegetation, snow cover and other surface conditions (Kurylyk & Hayashi, 2016). To capture the intra-year dynamics of the  $n$ -factor, we introduced timestamp ( $ts$ , consisting of month and day) and encoded it with Fourier transformation using an FC layer, which is a common practice in time-series neural networks such as Informer to transform timestamp from the time domain to the frequency domain and extract periodic patterns (H. Zhou et al., 2021). As for the volumetric water content of liquid water, in some studies, it is generally regarded as a constant that does not change with time due to the lack of long-term observations (Gao et al., 2022). And the thermal conductivity is often determined by Johansen's parameterization scheme and is considered to vary with soil water content (Qin et al., 2016). Once the volumetric water content of liquid water is determined as a constant, the thermal conductivity is also regarded to be constant (Gao et al., 2022; Johansen, 1977). However, these assumptions may not be reasonable in the context of significant climate change. Based on the aforementioned consideration, the soil water content ( $s$ ) output by the P-RNN layer (i.e., EXP-HYDRO model) is used to reflect the dynamics of the volumetric water content of liquid water. All these sequences, including the air temperature, embedded timestamp vector, and soil water content, are concatenated and then fed into the LSTM layer to learn the mechanisms of how the freezing depth impacts the fast flow ( $q_f$ ) and slow flow ( $q_s$ ) output by the P-RNN layer, where the impacts are quantified by two corrections  $k_f$  and  $k_s$ , respectively. The final  $Q^*$  is calculated by Equation (1) and fed into the subsequent pure data-driven part, which is the same two-layer Conv1D network as in the hybrid-J model.

$$Q^* = k_f \times q_f + k_s \times q_s \quad (1)$$

In Equation (1), the  $k_f$  and  $k_s$  represent the impact of freeze-thaw processes on runoff, with a value of one indicating no effect, values less than one indicating a reduction in runoff due to freeze-thaw, and values greater than one indicating an increase in runoff due to freeze-thaw. In order to ensure that  $k_f$  and  $k_s$  remain neutral around the value of 1 and not biased toward either effect, the “tanh+1” activation function, which is derived from the “tanh” activation function and centered at point (0, 1), is used to constrain the range of  $k_f$  and  $k_s$  to (0, 2). Previous simulations using a distributed model in the SRYR have shown that the relative increase in each runoff component due to freeze-thaw processes does not exceed two-fold (A. Sun et al., 2020), indicating that the value range of (0, 2) is reasonable.

It is worth noting that we do not explicitly calculate the freezing depth in the frozen soil module but take this implicit way to couple the forcing of the Stefan equation into the model. This is mainly due to the great uncertainty in the parameterization of the Stefan equation, including but not limited to the error caused by taking constant  $n$ -factor, volumetric water content of liquid water, and thermal conductivity, as mentioned above. Another consideration for this implicit coupling is to fully utilize the LSTM's powerful ability to simulate long-term hydrological processes. The core of the Stefan equation for calculating freezing and thawing indices lies in the integration of temperature, which bears a natural resemblance to the memory storage and updating mechanism of LSTM. At the same time, the underrepresented processes in the Stefan equation can also be effectively complemented by LSTM's strong learning capabilities. Recent studies have shown that the cell state in the trained rainfall-runoff LSTM strongly correlates with hydrological states, such as snow cover and soil water storage (Kratzert, Herrnegger, et al., 2019; Lees et al., 2022), demonstrating that the LSTM layer is especially capable of capturing these long-term system dynamics. This further inspires us to design an LSTM layer for learning the dynamics of soil freeze-thaw and its impacts on runoff rather than using the Stefan equation directly. When designing the structure of the frozen soil module, we aimed to ensure that the LSTM could effectively learn frozen soil dynamics while minimizing the introduction of additional parameters that may weaken the physical basis. Therefore, we opted for a relatively small encoding dimension for the periodic timestamp encoding and a small number of hidden layers for the LSTM. In order to distinguish the physics-informed DL model coupled with the frozen soil module from the hybrid-J model, in the following sections, we refer to our hybrid model as the “hybrid-Z model.” Based on the above development, it becomes evident that the most significant differentiation and innovation of the hybrid-Z model, as opposed to the hybrid-J model, is the inclusion of an implicitly coupled

frozen soil module. This module, which is based on the integration of the neural networks and EXP-HYDRO model, serves to account for the intricate process of soil freeze-thaw.

### 2.2.2. Other Types of Models for Comparison

The hybrid-J model is used as one of the models to compare with our newly developed hybrid-Z model to verify the effectiveness of the frozen soil module. In addition, a lumped model, a physically-based distributed model, and a standard DL model are selected respectively to compare with the physics-informed DL model to assess its ability to simulate runoff response to climate change.

Among them, the lumped model and the standard DL model are both components of the physics-informed DL model, which are the single P-RNN layer (i.e., the EXP-HYDRO model) and the purely data-driven part after substituting the P-RNN layer with an LSTM layer, respectively. A comparison of these two models with our hybrid-Z model will help enlighten the role of the data-driven components and infused physics in the physics-informed DL model, respectively. Besides, The distributed geomorphology-based eco-hydrological model (GBEHM) (B. Gao, Qin, et al., 2016), a physically-based model widely used in alpine catchments, especially the QTP, is selected as the representation of the distributed hydrological model to compare. The GBEHM model takes the hillside as the calculation unit and simulates the hydrological impact of soil freezing and thawing processes based on energy balance and temperature-based heat transfer equations. The performance of the GBEHM model has been verified in the source region of the Yellow River, Yangtze River, and Upper Heihe Basin (B. Gao, Qin, et al., 2016; Qin et al., 2017; Shi et al., 2020), which means it is well qualified for runoff simulation in alpine catchments. For more model information, please refer to B. Gao, Qin, et al. (2016).

## 2.3. Experiments Design

### 2.3.1. Model Establishment and Parameter Calibration

In order to assess model performance in watersheds with various permafrost area ratios, we separately built the hybrid-Z model in each of the three subbasins in the SRYR. The observed runoff from the TNH, MAQ, and JIM stations was then used to train and verify the model, respectively. And the models for comparison, including the physics-informed DL model without soil freeze-thaw process (i.e., the hybrid-J model), EXP-HYDRO model, and standard DL model, were all separately built in the three subbasins in the same way. The daily basin-average meteorological data were used to force the four models mentioned above, among which the inputs of the hybrid-Z model, hybrid-J model, and standard DL model were the complete six climatic data and timestamp, and those of the EXP-HYDRO model were precipitation, air temperature, and potential evapotranspiration. The four models were developed using the PyTorch framework, and the calibration and evaluation of the models strictly adhered to the standard procedures of DL models. The whole data were divided into the training set, validation set, and test set, and the parameters were automatically determined based on the gradient descent method. When subdividing the data into three parts, we used the first 30 years for training and validation and the second 30 years for testing. This is based on the fact that the climate in the first 30 years was relatively stable before 1990 but changed rapidly after 1990, which can be referred to in Section 3.1 and relevant studies (P. Wu, Liang, et al., 2020). Therefore, such a data partitioning method can fully test the stability of DL-related models under climate change.

In addition, the GBEHM model was also built to provide a thorough comparison. We established a GBEHM model with a 3-km spatial resolution and a 1-hr time resolution. A 50-year warm-up run was used to initialize the hydrological conditions. The daily streamflow data during 1981–1986 at the TNH station were used to manually calibrate the model parameters, and the runoff from the MAQ and JIM stations was used for verification. After the calibration, the model was applied in the entire simulation period from 1965 to 2015. For more model settings, including the interpolation of daily meteorological data to the hourly scale and what parameters were selected to be calibrated, please refer to Qin et al. (2017).

During the model training (or calibration) process, the Nash-Sutcliffe efficiency coefficient (NSE) (Nash & Sutcliffe, 1970) (Equation 2) was chosen as the loss function. Given that the NSE is more sensitive to the model's simulation of the peaks (Gupta et al., 2009), we also took into account the performance of the model during the baseflow period when designing the loss function. Additionally, the dominant factors for runoff response vary across time scales, so the model's performance at the long-term scale, which in this case is the annual scale, was also considered in the loss function to guide the model (especially DL-related models) to learn the different

underlying runoff mechanisms better. Therefore, the final loss function is calculated by NSE weighting at different time scales, as shown in Equation 3.

$$NSE = 1 - \frac{\sum_{t=1}^n (Q_o^t - Q_s^t)^2}{\sum_{t=1}^n (Q_o^t - \overline{Q_o})^2} \quad (2)$$

where  $Q_o$  is observed streamflow,  $\overline{Q_o}$  is the observed average runoff, and  $Q_s$  is simulated streamflow.

$$\text{loss} = 1 - (\phi_1 \times NSE_{\text{daily}} + \phi_2 \times NSE_{\text{baseflow}} + \phi_3 \times NSE_{\text{annual}}) \quad (3)$$

where  $NSE_{\text{daily}}$ ,  $NSE_{\text{baseflow}}$ , and  $NSE_{\text{annual}}$  are the NSE for the daily scale, baseflow period, and annual scale. The baseflow period of each station was determined based on the study of T. Zhang et al. (2022) and P. Wu, Liang, et al. (2020). Specifically, at the TNH and MAQ stations, the baseflow from January to February was dominant, while at the JIM station, it was from January to March.  $\phi_1$ ,  $\phi_2$ , and  $\phi_3$  are the weight at three scales, respectively, which are hyper-parameters determined by training, and  $\sum_{i=1}^3 \phi_i = 1$ . For more details on hyperparameter tuning, please refer to Text S2 in Supporting Information S1.

### 2.3.2. Model Comparison Schemes and Evaluation Metrics

Scheme 1: in order to evaluate the physics-informed DL model in simulating runoff, the performance of the hybrid-Z model and hybrid-J model is compared.

If the frozen soil module is effective, there should be an improvement in runoff simulation. Therefore, in this section, we mainly focus on comparing the two models' differences in simulating runoff and the impacts of soil freeze-thaw on runoff.

Regarding the evaluation of runoff simulation, it should be noted that the sensitivity of model parameters varies with runoff at different time scales (H. Xie et al., 2017), so a good performance of the model on runoff simulation at a fine time scale does not necessarily imply that it is good at a coarse time scale, and vice versa (Setti et al., 2022). The model's simulation of streamflow at different time scales reflects its capacity to capture the different runoff responses to climate and the underlying surface. So, in this paper, the models' performance in simulating daily, monthly, and annual runoff will be compared in detail to reflect their ability to capture hydrological responses at different timescales. In addition to the NSE, the Pearson correlation coefficient ( $r$ ) and Percent bias (PBIAS) are also used to evaluate the model's performance on runoff. The metrics are calculated as follows:

$$r = \frac{\sum_{t=1}^n (Q_o^t - \overline{Q_o})(Q_s^t - \overline{Q_s})}{\sqrt{\sum_{t=1}^n (Q_o^t - \overline{Q_o})^2} \sqrt{\sum_{t=1}^n (Q_s^t - \overline{Q_s})^2}} \quad (4)$$

$$PBIAS = \frac{\sum_{t=1}^n (Q_s^t - Q_o^t)}{\sum_{t=1}^n Q_o^t} \times 100\% \quad (5)$$

where the subscript “o” denotes the observed value, and “s” denotes the simulated value.

As introduced in Section 1, the impacts of frozen soil on runoff are easily masked by other factors, so the performance of runoff simulation is insufficient to fully depict the model's capacity to simulate the impacts of soil freeze-thaw on runoff. Given that a large number of studies have shown that the impacts of the soil freeze-thaw process on runoff can be mainly reflected in the hydrography recession (Gao et al., 2021), the  $Q_{\text{max}}/Q_{\text{min}}$  ratio (Ye et al., 2009) and the discontinuous baseflow recession phenomenon (DBR) (Gao et al., 2022) were selected to evaluate the performance of the model in simulating the impacts of soil freeze-thaw on runoff from different aspects. The  $Q_{\text{max}}/Q_{\text{min}}$  ratio, initially defined as the ratio of maximum monthly runoff to minimum monthly runoff and later expanded to include the ratio of summer rainfall flow to winter baseflow when applied in the SRYR (P. Wu, Liang, et al., 2020), was widely used to quantify the impact of permafrost degradation on runoff (Gao et al., 2021; L. Liu, Zhang, et al., 2020). The  $Q_{\text{max}}/Q_{\text{min}}$  ratio is mainly based on the understanding that permafrost degradation increases soil infiltration and percolation during the thawing summer season, enhancing groundwater storage, which in turn leads to the decline of summer rainfall flow and the rise of winter baseflow (A. Sun et al., 2020; Walvoord & Kurylyk, 2016; T. Zhang et al., 2022). Compared with the  $Q_{\text{max}}/Q_{\text{min}}$  ratio, which reflects the impact of permafrost degradation cumulatively,

the DBR reflects the microscopic process of soil freeze-thaw affecting the hydrograph. It refers to a discontinuous baseflow recession process caused by the dramatic disturbance in the groundwater system when the water in the supra-permafrost layer is frozen, and then the recession is only contributed by the seasonal frozen soil area. In the SRYR, permafrost is mostly distributed in the basin above the JIM station and the permafrost degradation is more significant there, while the downstream basin is mainly dominated by seasonally frozen ground (Figure 1c). Therefore, we only consider the  $Q_{\max}/Q_{\min}$  ratio and DBR phenomenon at the JIM station when comparing the model's simulation of the impacts of soil freeze-thaw on runoff. According to P. Wu, Liang, et al. (2020), the  $Q_{\max}/Q_{\min}$  ratio was calculated as the ratio of summer rainfall flow to winter baseflow. The rainfall flow and baseflow are determined by the respective dominant periods. In detail, rainfall flow dominates runoff from July to September and hence is calculated as the average runoff of these 3 months, and similarly, the baseflow is calculated as the average runoff from January to March (T. Zhang et al., 2022). The model's simulation of the  $Q_{\max}/Q_{\min}$  ratio was evaluated using the NSE.

Scheme 2: In order to comprehensively verify the ability of the physics-informed DL model to simulate runoff response to climate change, the performance of the hybrid-Z model is compared with other types of models, including the lumped model (EXP-HYDRO), physically-based distributed model (GBEHM), and standard DL model.

In Scheme 2, the two models' performance in simulating streamflow at multi-timescales at the three stations and the  $Q_{\max}/Q_{\min}$  ratio at the JIM station are compared first. Additionally, the responses of the simulated annual streamflow and the  $Q_{\max}/Q_{\min}$  ratio to climate change were also calculated to compare with the observation in order to assess the model's ability to simulate runoff change under warming. Here, the responses of the annual streamflow and the  $Q_{\max}/Q_{\min}$  ratio to climate change are quantified by the relative change and trend. The relative change was calculated as the difference of the mean annual streamflow or the  $Q_{\max}/Q_{\min}$  ratio between the rapid change period (after 1990) and the slow change period (before 1990). And the trend is calculated over the entire period using the Mann–Kendall (M-K) trend test (Gilbert, 1987; Kendall, 1948; Mann, 1945). The details of the M-K trend test will be introduced in Section 2.4. The relative change and trend reflect different aspects of runoff variation and can thoroughly test whether the model can replicate runoff changes in the changing climate.

We use the relative error (RE) to quantify the deviation between the simulated responses and observed responses of annual streamflow (or the  $Q_{\max}/Q_{\min}$  ratio) to climate change. The RE is calculated by the following equation:

$$RE = \left| \frac{x_s - x_o}{x_o} \right| \times 100\% \quad (6)$$

where the  $x_s$  is the relative change (or trend) of the simulated value, and  $x_o$  is the relative change (or trend) of the observed value.

Comprehensively evaluating the performance of the models involves many different evaluation metrics (16 metrics for one model, including the NSE,  $r$ , and PBIAS for multi-timescale streamflow at the three stations, the NSE for the  $Q_{\max}/Q_{\min}$  ratio at the JIM station, and the RE for the responses of the annual streamflow at the three stations and the  $Q_{\max}/Q_{\min}$  ratio at the JIM station). It is difficult to have a model that performs best on all evaluation metrics in all three subbasins, so we used a score-based method (Fu et al., 2013; Lun et al., 2021) to give a comprehensive evaluation of the models. The rank score (RS) is calculated as follows:

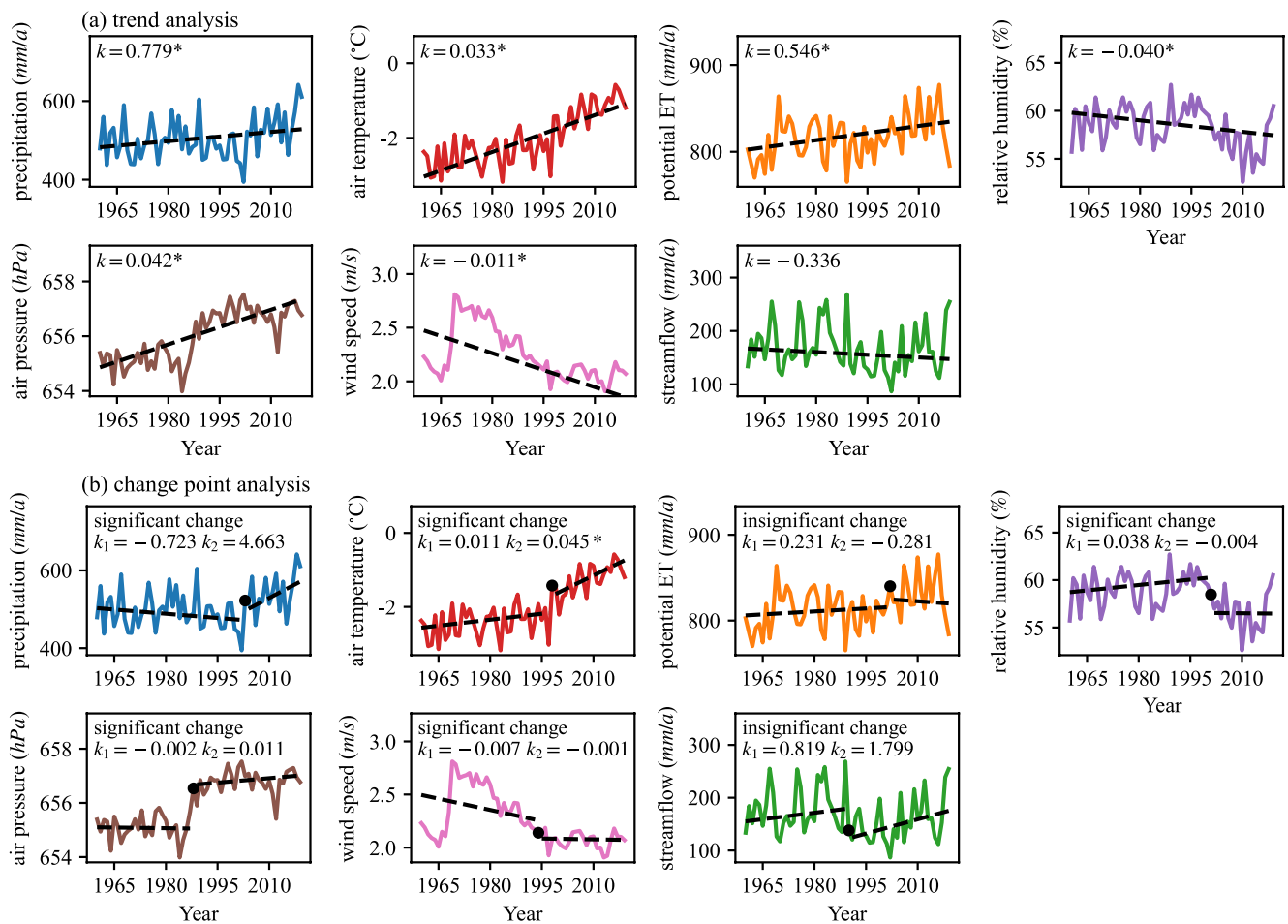
$$R_i = \begin{cases} \frac{X_i - X_{\min}}{X_{\max} - X_{\min}} \times 10 & \text{if the model is better with a higher } X \\ \frac{X_{\max} - X_i}{X_{\max} - X_{\min}} \times 10 & \text{if the model is better with a lower } X \end{cases} \quad (7)$$

$$RS = \frac{1}{n} \sum_{i=1}^n R_i \quad (8)$$

where  $n$  is the number of evaluation metrics,  $X_i$  is the RE (or other evaluation metrics such as the NSE,  $r$ , and PBIAS),  $X_{\max}$  and  $X_{\min}$ , respectively is the maximum and minimum RE (or other evaluation metrics) of all models. The RS ranges from 0 to 10, with larger values indicating better model performance.

## 2.4. Statistical Methods

Two non-parametric tests, the Mann–Kendall (M-K) trend test (Gilbert, 1987; Kendall, 1948; Mann, 1945) and the Pettitt change point test (Pettitt, 1979), were used to analyze the trend and change points of climatic data. More details about the two tests could be found in Text S3 in Supporting Information S1.



**Figure 3.** The results of the M-K trend test and the Pettitt change point test of the climate in the SRYR and the streamflow at the TNH station. (a) The trend analysis. The solid line is the observation, and the dashed line represents the trend. (b) The change point analysis. The point indicates when the abrupt change occurs. The \* means the trend is significant with  $p < 0.05$ .

### 3. Results

#### 3.1. Trend and Change Point of the Climatic Factors in the SRYR

As shown in Figure 3, from 1960 to 2019, the climate of the SRYR has changed significantly, in which the annual precipitation, temperature, and potential evapotranspiration have a significant increasing trend with 0.779 mm, 0.033°C, and 0.546 mm per year, respectively, and the annual relative humidity, wind speed, and air pressure have also changed with a significant trend. According to the Pettitt change point test results, all climate variables, except for the potential evapotranspiration, had an apparently abrupt change around the 1990s. As seen, the annual precipitation and temperature showed a relatively weak trend of  $-0.723$  mm and  $0.011^{\circ}\text{C}$ , respectively, before the 1990s. While the absolute trends following the 1990s are six and four times greater than before, and the difference in the mean annual temperature between the two periods was as high as  $1^{\circ}\text{C}$ . The results fully indicate that the climate in the SRYR has undergone rapid changes, and these changes have intensified in the last 30 years. And we also noticed that, in the face of such significantly changing climate, both the trend and abrupt change of the observed streamflow are non-significant, which may be attributed to the competing effects of meteorological factors on various hydrological processes. For example, the surface water waste brought about by increased evapotranspiration and infiltration may cancel out the subsurface flow recharge caused by frozen soil degradation, as mentioned above. These findings underscore the complexity of runoff regimes in this region and the imperative to leverage DL techniques to enhance runoff simulation.

Based on the above results, we divide the 60 years from 1960 to 2019 into two periods: the slow change period (1960–1989) and the rapid change period (1990–2019). As mentioned in Section 2.3.1, in order to rigorously

verify the stability and consistency of the models under climate change, we chose the slow change period as the training and validation set and the rapid change period as the test set. In detail, the periods of the training set, validation set, and test set were 1960–1984, 1985–1989, and 1990–2019, respectively.

### 3.2. Evaluation of the Physics-Informed DL Model in Simulating Runoff

#### 3.2.1. The Two Models' Performance in Simulating Streamflow at Multi-Timescales

This study built the separate hybrid-Z model and hybrid-J model for three subbasins from 1960 to 2019. Figure 4 presents the performance of two models on the test set at the daily scale, monthly scale, and annual scale, respectively. For a more intuitive comparison, Table 2 shows the differences between the two models on the test set in particular.

In general, the hybrid-Z model and the hybrid-J model exhibit comparable performance for the TNH and JIM stations, although the former exhibits a slight advantage in terms of the NSE at the annual scale. However, in the basin above the JIM station, this advantage is particularly significant—the NSE of the hybrid-Z model is 0.82, while that of the hybrid-J model is 0.71. Additionally, the hybrid-Z model also demonstrates significantly better performance than the hybrid-J model in terms of PBIAS (−0.67% compared to −6.95%). As seen in Figure 4, the simulated annual runoff of the hybrid-J model at the JIM station is apparently lower than the observation, especially for high annual flows, while the simulations of the hybrid-Z model are evenly distributed on both sides of the 1:1 line. The significant advantage of the hybrid-Z model at the JIM station, where the permafrost degradation is more significant, to some extent reflects the effectiveness of the addition of frozen soil module. Specifically, the superior NSE of the hybrid-Z model at the annual scale may be due to its better ability to capture some inter-annual responses, and the smaller PBIAS demonstrates the hybrid-Z model's superiority in simulating water balance.

#### 3.2.2. Evaluation of the Physics-Informed DL Model in Simulating the Impacts of Soil Freeze-Thaw on Runoff

As mentioned above, the  $Q_{\max}/Q_{\min}$  ratio characterizes the hydrograph recession property, which is closely related to permafrost degradation. The two models' simulations of the  $Q_{\max}/Q_{\min}$  ratio at the JIM station are shown in Figure 5. Although the NSE of the hybrid-J model and hybrid-Z model are not very ideal, this is mainly due to the difficulty of baseflow simulation caused by the low signal-to-noise ratio for low-flow observation. Comparatively, the hybrid-Z model performs slightly better than the hybrid-J model, with the NSE increasing by 0.18.

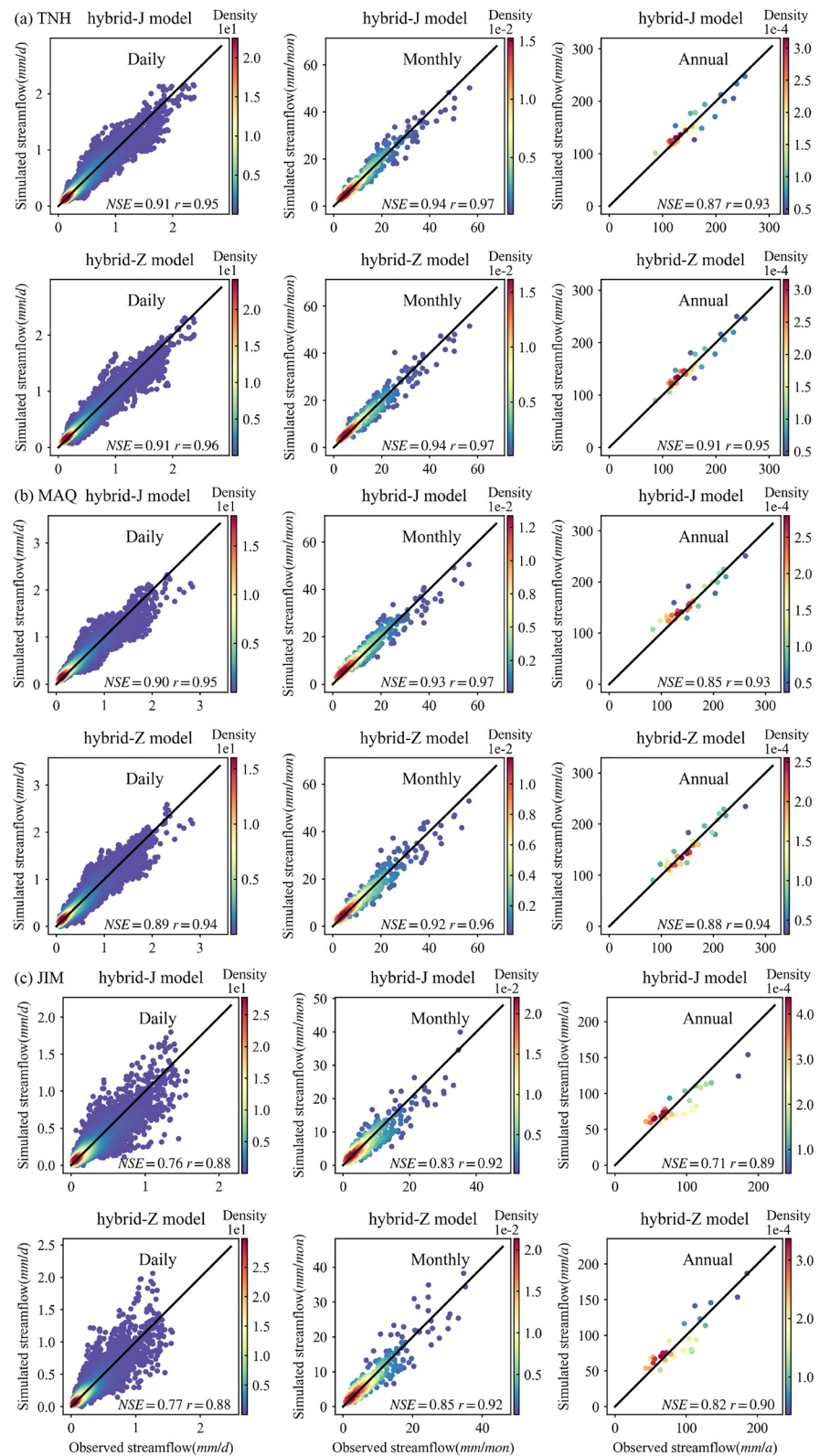
As for the simulation of the DBR event, the runoff at the JIM station in the training set and test set was selected for analysis respectively (Figures 6a and 7a). The hydrograph shows a distinct discontinuous baseflow recession each year, with the turning point typically occurring in mid-to late-December. After the turning point, runoff declines at a faster rate than before because the groundwater beneath the permafrost is mostly frozen, causing a reduction in groundwater discharge. Consistent with the observed runoff, the bending down discontinuous baseflow recession phenomenon also occurs in the simulated runoff of the hybrid-Z model (Figures 6c and 7b). Although the speed of the recession in the two periods may not be exactly the same as the observation, we still believe that the hybrid-Z model can effectively simulate the DBR considering the difficulty of perfectly reproducing such a microscopic process. The hybrid-J model, however, is unable to reproduce this phenomenon perfectly, and there are many sudden spikes in the hydrograph, which meant that hybrid-J model is unstable in the simulation of low flow (Figures 6b and 7b).

In general, the hybrid-Z model demonstrates better performance compared to the hybrid-J model in simulating both the  $Q_{\max}/Q_{\min}$  ratio and DBR phenomenon. In addition, the dynamics of LSTM's memory cells, as well as  $k_f$  and  $k_s$  are visualized in Figure S1 in Supporting Information S1. The results show that the memory cells can capture the threshold temperature of 0°C and activate different neurons for state updates, and  $k_f$  and  $k_s$  exhibit seasonal fluctuations consistent with freeze-thaw in the three sub-basins, especially for  $k_s$ , which is in agreement with our basic understanding of freeze-thaw. This demonstrates that the incorporation of the physics related to soil freeze-thaw can better guide the physics-based DL model to learn the impacts of the soil freeze-thaw process on runoff.

### 3.3. Comparison of the Physics-Informed DL Model With Other Types of Models

#### 3.3.1. The Four Models' Simulations of the Multi-Timescale Streamflow and the Impacts of Soil Freeze-Thaw on Runoff

The four models' performance of streamflow simulation at the TNH, MAQ, and JIM station on the test set (or the verification period for the GBEHM) are shown in Figures S2–S4 in Supporting Information S1,



**Figure 4.** Comparison of simulated and observed streamflow of the two models at daily, monthly, and annual scales on the test set. (a), (b) and (c) are the results at the TNH, MAQ, and JIM stations, respectively.

**Table 2**  
The Streamflow Simulation Results of the Two Models on the Test Set

| Station | Model    | Daily |          | Monthly |          | Annual |          | PBIAS (%) |
|---------|----------|-------|----------|---------|----------|--------|----------|-----------|
|         |          | NSE   | <i>r</i> | NSE     | <i>r</i> | NSE    | <i>r</i> |           |
| TNH     | Hybrid-J | 0.91  | 0.95     | 0.94    | 0.97     | 0.87   | 0.93     | 0.20      |
|         | Hybrid-Z | 0.91  | 0.96     | 0.94    | 0.97     | 0.91   | 0.95     | 0.82      |
| MAQ     | Hybrid-J | 0.90  | 0.95     | 0.93    | 0.97     | 0.85   | 0.93     | 3.04      |
|         | Hybrid-Z | 0.89  | 0.94     | 0.92    | 0.96     | 0.88   | 0.94     | −0.86     |
| JIM     | Hybrid-J | 0.76  | 0.88     | 0.83    | 0.92     | 0.71   | 0.89     | −6.95     |
|         | Hybrid-Z | 0.77  | 0.88     | 0.85    | 0.92     | 0.82   | 0.90     | −0.67     |

Note. The PBIAS is a metric that accumulates over time, so its value is constant across scales.

respectively, and a more intuitive summary of the performance was presented in Table 3. Clearly, the hybrid-Z model achieves the best performance at all stations and on all time scales. The standard DL model can well simulate the streamflow at the daily and monthly scales but performs less well at the annual scale, which may reveal the limitations of purely data-driven models in the complex and changing climate. That is, they may not be able to learn the inter-annual responses due to the lack of crucial physics. In addition, the GBEHM model works well at the downstream TNH and MAQ stations but fails in the upstream JIM station.

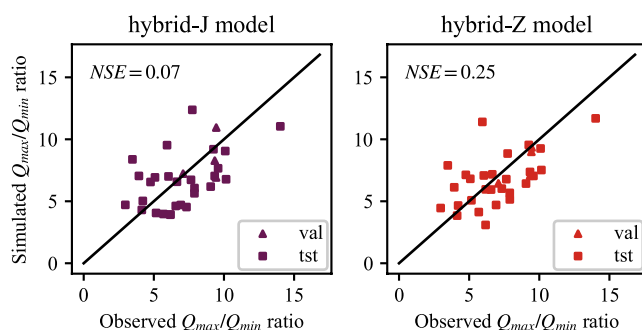
The four models' simulations of the  $Q_{\max}/Q_{\min}$  ratio from 1985 to 2015 are presented in Figure 8. Compared with other models, the physics-informed DL model performs best in terms of simulating the impacts of soil freeze-thaw on runoff, followed by the GBEHM model. The standard DL model without physics performs poorly, further illustrating the importance of infusing physics into DL.

### 3.3.2. The Four Models' Ability to Simulate Runoff Response to Climate Change

In order to assess whether the model can replicate runoff changes in response to climate change, we first analyzed the ability of the four models to simulate the relative change and trend of annual streamflow from 1965 to 2015, which is shown in Figures 9 and 10. And the REs between the simulated and observed relative change and trend were also calculated in Table 4. In terms of simulating the relative change of annual mean runoff, the GBEHM and the hybrid-Z model performs comparably. And for the trend simulation, the hybrid-Z model performs best, followed by the GBEHM. It is worth mentioning that the EXP-HYDRO model and standard DL model perform almost the worst in the relative change and trend simulation, but their combination (i.e., the hybrid-Z model) is well capable of predicting annual runoff under climate change. This once again emphasizes the great potential of the combination of DL and physics.

Similarly, the relative change and trend of the  $Q_{\max}/Q_{\min}$  ratio were analyzed, as shown in Figure 11, and the RE for each metric was presented in Table 4. Consistent with previous studies, we found that the observed  $Q_{\max}/Q_{\min}$  ratio at the JIM station has been decreasing with permafrost degradation. And the decrease in the  $Q_{\max}/Q_{\min}$  ratio is well reproduced by the hybrid-Z model, with a relative change of −2.2 and a trend of −0.6, which is very close to the observed −1.3 and −0.5. In contrast, the GBEHM and standard DL models apparently overestimate this decline, and the EXP-HYDRO model fails to even capture the decline, with an exact opposite trend.

Last but not least, the RS is calculated in order to evaluate the models' overall performance, as shown in Table 5. The result shows that the EXP-HYDRO model, standard DL model, and GBEHM model are not well qualified, with RS of 2.65, 4.16, and 6.03, respectively. In contrast, the hybrid-Z model can achieve the top performance on almost all the simulations, indicating that the physics-informed DL model is competent for runoff simulation in alpine catchments under shifting domain conditions caused by climate change.

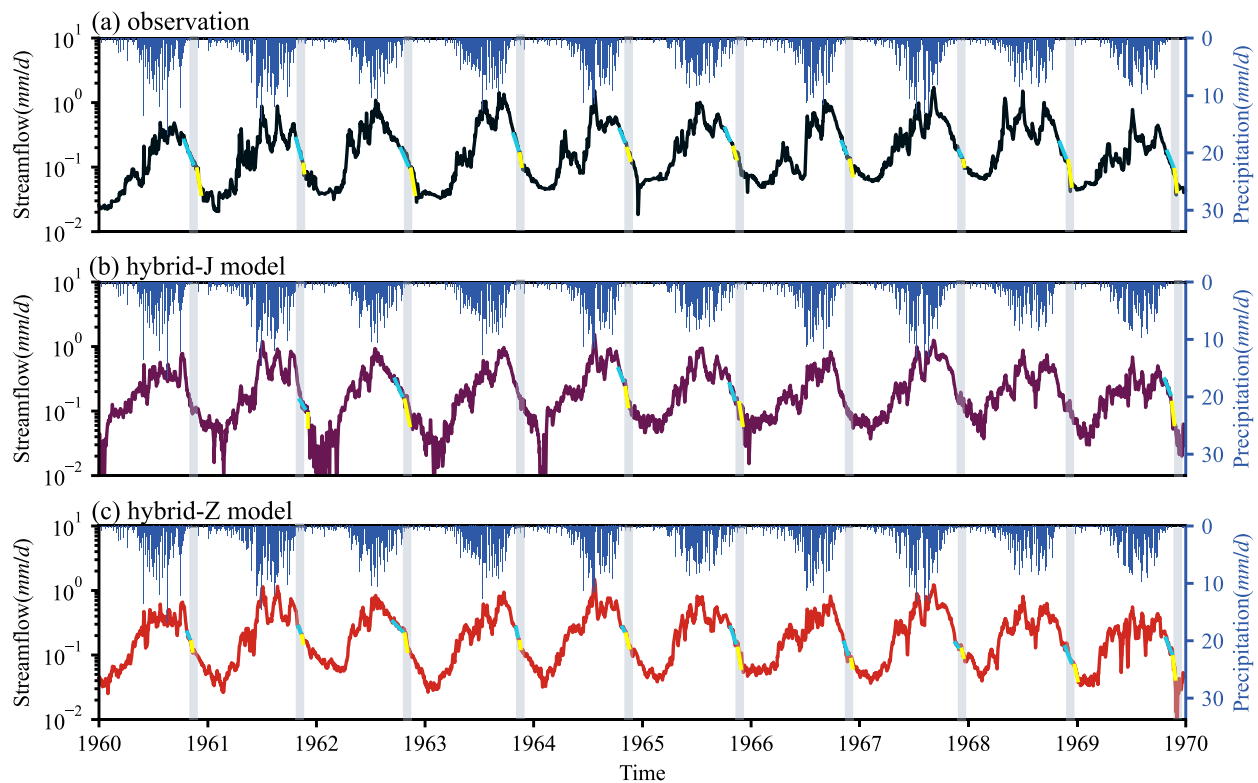


**Figure 5.** The performance of the two models in simulating the  $Q_{\max}/Q_{\min}$  ratio at the JIM station during 1985–2019. The “val” and “tst” mean validation set and test set, respectively.

## 4. Discussions

### 4.1. The Infused Physics May Make DL Applicable in the Climate Change Projection

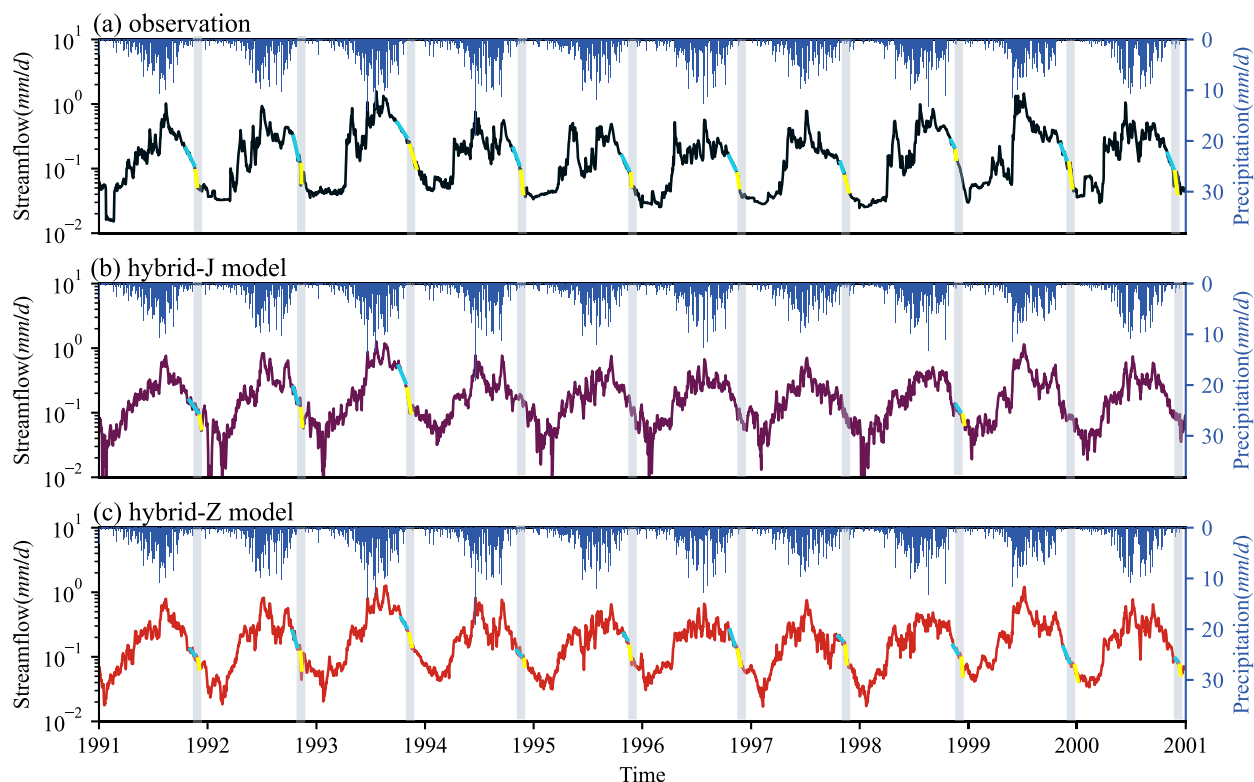
Runoff projection under future climate is a common task in hydrological modeling applications, which refers to predicting how a system will respond to possible future changes in climate or the underlying surface. However, the reliability of DL in climate change projection has been doubted by hydrologists because it may require DL models to extrapolate to scenarios unseen in the training set (Razavi et al., 2022). Although some studies have shown that LSTM trained on large-sample basins like CAMELS can exhibit greater stability under climate change (Wi & Steinschneider, 2022), the feasibility of applying LSTM models trained on large samples to the SRYR remains to be verified, given that the proportion of alpine catchments in these datasets is



**Figure 6.** The DBR phenomena at the JIM station in the training set. The transparent bars indicate when the DBR event occurs. (a) Observation. (b) The hybrid-J model. (c) The hybrid-Z model. The cyan and yellow lines represent changes in the velocity of groundwater recession.

relatively low (with only 2.8% of catchments situated above 3,000 m), and the hydrological processes in alpine catchments are unique and complex. Therefore, the focus of this paper is how to improve the applicability of DL models under climate change, particularly in the absence of large-sample model training. In our study, we found that when physics is infused into DL, the consistency of the physics-informed DL model in extrapolation predictions is greatly enhanced. In the SRYR, the climate of the two periods changed rapidly, with average temperature differing by about 1°C. This presents a substantial difficulty for DL because the distribution of the data on the test set differs significantly from that on the training set, potentially making the underlying regularities in the data learned by DL inapplicable. This view is supported to some extent by the disparity in the performance of standard DL model at annual streamflow and the failure of the simulations of  $Q_{\max}/Q_{\min}$  ratio. The physics-informed DL model's performance, however, does not appear to be impacted by the altered data distribution. The NSEs of multi-timescale streamflow simulation at the TNH and MAQ stations are around 0.90, and even in the basin above the JIM, where permafrost degradation is more significant, the NSE of daily runoff is 0.77, and the NSEs of monthly and annual runoff are above 0.8. Moreover, the physics-informed DL can also accurately capture how runoff changes in response to climate change.

In order to further highlight the role of the physics of dominant hydrological processes on model consistency, we calculated the relative deterioration of the three models' NSE on the test set versus the validation set (Table 6). A small relative deviation means that the model has strong consistency when making the out-of-sample prediction. Overall, the performance gap between the testing and validation sets is smaller for the hybrid-Z model than for the hybrid-J model, and smaller than for the standard DL model, especially at the annual scale. Compared with the standard DL model, the hybrid-J model has more physics related to runoff generation, as well as snow accumulation and melting processes hidden in the P-RNN layer. And in comparison with the hybrid-J model, the hybrid-Z model additionally takes into account soil freeze-thaw and therefore better reflects the basin-dominated hydrological processes. So, the varying degrees of physics incorporated into DL are respectively reflected in the three models. In this regard, Table 6 shows an interesting result that the consistency of DL is better enhanced as dominant hydrological processes in the basin are more considered in the infused physics.



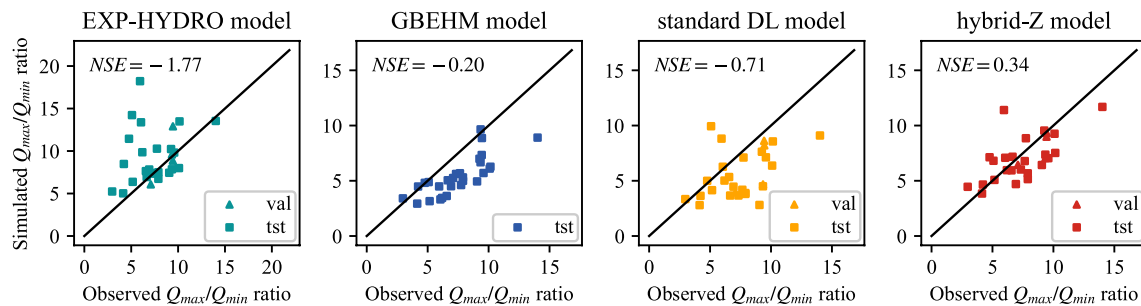
**Figure 7.** The DBR phenomena at the JIM station in the test set. The transparent bars indicate when the DBR event occurs. (a) Observation. (b) The hybrid-J model. (c) The hybrid-Z model. The cyan and yellow lines represent changes in the velocity of groundwater recession.

So why can the physics-informed DL model achieve better performance in extrapolation than the standard DL model? A plausible explanation is that the incorporation of physics aids DL's understanding of the fundamentals of the dominant watershed hydrology processes, including runoff generation, snow accumulation and melting, as well as the soil freeze-thaw process. Our discovery can be further supported by earlier research. For instance, S. Jiang, Zheng, Yi, et al. (2020) output the learned snowpack ( $S_0$ ) in the P-RNN layer and found the dynamic of  $S_0$  perfectly

**Table 3**  
The Multi-Timescale Streamflow Simulation of the Four Models on the Test Set

| Station | Model       | Daily       |             | Monthly     |             | Annual      |             | PBIAS(%)     |
|---------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|
|         |             | NSE         | $r$         | NSE         | $r$         | NSE         | $r$         |              |
| TNH     | EXP-HYDRO   | 0.81        | 0.92        | 0.86        | 0.95        | 0.62        | 0.90        | 10.52        |
|         | GBEHM       | 0.73        | 0.87        | 0.91        | 0.96        | 0.88        | 0.95        | -4.50        |
|         | Standard DL | 0.90        | 0.95        | 0.93        | <b>0.97</b> | 0.82        | 0.93        | 6.16         |
|         | Hybrid-Z    | <b>0.91</b> | <b>0.96</b> | <b>0.94</b> | <b>0.97</b> | <b>0.91</b> | <b>0.95</b> | <b>0.82</b>  |
| MAQ     | EXP-HYDRO   | 0.80        | 0.91        | 0.85        | 0.93        | 0.63        | 0.90        | 10.14        |
|         | GBEHM       | 0.74        | 0.87        | 0.88        | 0.94        | 0.79        | 0.92        | 3.68         |
|         | Standard DL | 0.86        | 0.93        | 0.88        | 0.95        | 0.84        | 0.93        | -8.78        |
|         | Hybrid-Z    | <b>0.89</b> | <b>0.94</b> | <b>0.92</b> | <b>0.96</b> | <b>0.88</b> | <b>0.94</b> | <b>-0.86</b> |
| JIM     | EXP-HYDRO   | 0.64        | 0.86        | 0.70        | 0.89        | 0.68        | 0.83        | 4.58         |
|         | GBEHM       | 0.01        | 0.83        | 0.31        | 0.90        | -0.82       | 0.84        | 42.68        |
|         | Standard DL | 0.73        | 0.86        | 0.78        | 0.89        | 0.59        | 0.77        | -2.77        |
|         | Hybrid-Z    | <b>0.77</b> | <b>0.88</b> | <b>0.85</b> | <b>0.92</b> | <b>0.82</b> | <b>0.90</b> | <b>-0.67</b> |

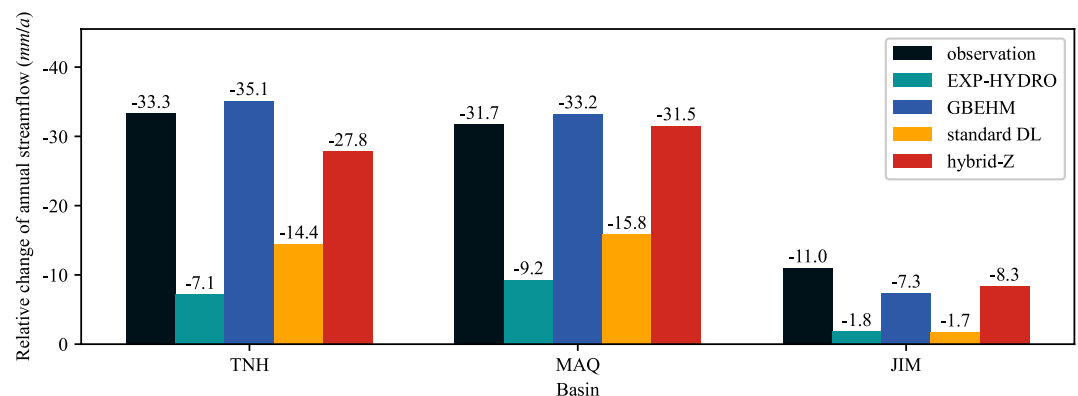
Note. Best performers under each metric are highlighted.



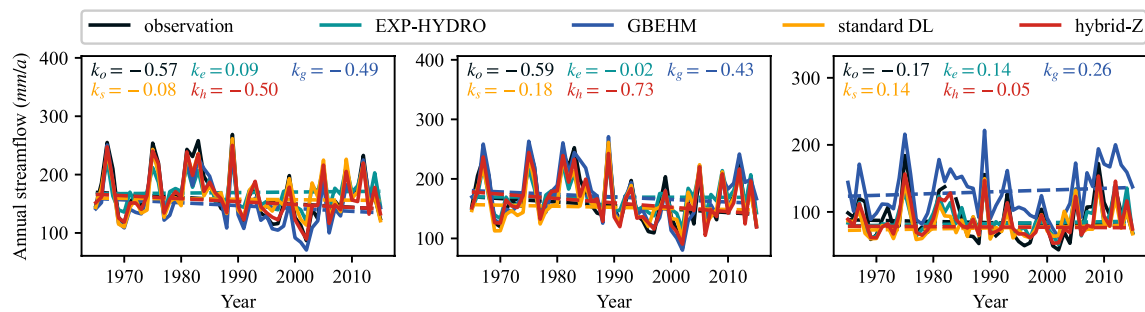
**Figure 8.** The performance of the Four models on the simulation of the  $Q_{\max}/Q_{\min}$  ratio at the JIM station during 1985–2015.

matched the daily snow water equivalent (SWE) dataset, with a correlation coefficient of 0.99. This demonstrated that the physics-informed DL model could accurately capture the dynamics of snow accumulation and melting under the guidance of the hybridization structure. Read et al. (2019) created a process-guided LSTM model to predict lake temperature by imposing a penalty for violating energy conservation in the loss function, in which case the energy balance is the dominant process influencing lake temperature. They found that the process-guided LSTM outperforms the purely data-driven model and process-based model in most cases when applied to instances outside the bounds of the training set. Zhao et al. (2019) integrated an ANN and a modified Penman-Monteith evapotranspiration equation, which includes the physics of dominant energy and water processes influencing potential evapotranspiration, to predict latent heat fluxes and compared the hybrid model with a standard DL model. They reported that the hybrid model is capable of conserving the surface energy balance while the standard DL model is not, so the former performs better in the out-of-sample prediction. Additionally, Razavi (2021) established a standard LSTM model and process-informed LSTM by adding a temperature threshold to separate the precipitation into rain and snow to simulate runoff. When applied to the simulation under the 2° C warming scenario, it turned out that the process-informed LSTM makes more consistent predictions with the physical model, but the standard LSTM exhibits a noticeable shift in peak flow timing. Similar experiment designs and results were also reported by Wi and Steinschneider (2022). They found that the LSTM model is not physically consistent under significant warming, and the model can make more realistic predictions when trained with evapotranspiration from SAC-SMA hydrological model as an additional input, but this does not hold for VIC or HYMOD evapotranspiration since the evapotranspiration estimates from the two models are not reliable due to the possible incorrect representations of soil moisture fluxes. So, the authors concluded that the hybrid framework might enhance DL's usability for hydrological projections under climate change, depending on the fidelity of physical models. In our study, the difference in the physical consistency of the hybrid-J model and hybrid-Z model with varying infused physics also demonstrates this point. That is, only when the infused physics can adequately characterize the dominant hydrological processes in the basin can DL make more reliable projections under climate change.

Naturally, another key issue surfaced, namely how to infuse physics into the DL to improve DL's reliability under climate change. Encoding a physical model as one component of the DL model is one of the common approaches



**Figure 9.** The relative change of simulated and observed annual streamflow between the rapid change period and slow change period.



**Figure 10.** The trend of annual simulated and observed streamflow during 1965–2015. (a) TNH. (b) MAQ. (c) JIM. Note that the value of  $k$  represents the trend, and the subscript corresponds to the first letter of the model's name.

(S. Jiang, Zheng, Yi, et al., 2020; Kraft et al., 2021; Zhao et al., 2019). In this study, we introduced the physics of runoff generation and snow accumulation and melting process into DL by considering the EXP-HYDRO model. Nevertheless, this approach explicitly incorporates hydrological processes wrapped in the physical model into DL and is only applicable when the understanding of these processes is sufficiently clear. However, there are some hydrological processes with knowledge gaps that make it challenging to write analytical equations explicitly, such as the soil freeze-thaw process, where the above method may be limited. In order to address this problem, we take an implicit form to couple the Stefan equation into DL. In detail, the Stefan equation provides expert knowledge that freezing depth is controlled by temperature, time, and soil water content. Given that the impacts of soil freeze-thaw on runoff at the watershed scale are complex and not fully understood, then, an LSTM network, which is fed with air temperature, time, and soil water content, is designed to learn the mechanisms of soil freeze-thaw affecting fast and slow flows. In this design, expert knowledge serves as the foundation for the neutral network architecture, and the data-driven component makes up for the lack of understanding of the frozen soil process. We also compared such an implicitly coupled frozen soil module with the architecture which is explicitly fed the Stefan freezing and thawing depth (for more information, please refer to Figures S5–S7 in Supporting Information S1). The explicitly coupled architecture's unsatisfactory performance in simulating the impacts of soil freeze-thaw on runoff further confirms the uncertainty brought by the parameterization of the Stefan equation, further highlighting the advantages of the implicit coupling in learning unrecognized processes. Similar implicit designs have also been shown in other studies. For instance, Li et al. (2022) realized that different underlying mechanisms of streamflow dynamics would lead to distinct behaviors in terms of flow magnitude and then used a process-based baseflow separation method to separate the streamflow process into two sub-processes, each of which was simulated by a data-driven model. The results show that the proposed model is better able to capture both high- and low-flow dynamics than purely data-driven models. Implicit coupling, it should be noted, is more in line with what hydrologists expect DL to do in the field of hydrology, namely employing DL's powerful learning capabilities to assist in understanding intricate hydrological processes with knowledge gaps. This design enables the data-driven model to learn specific hydrological processes under the guidance and constraints of expert knowledge, such as the impacts of soil freeze-thaw on runoff and the underlying mechanism of different flow magnitudes. In this way, physical and DL are more closely integrated. This further provides the possibility that, with the development of interpretable DL, the decision-making process in DL can be interpreted to help scientific discovery. Nevertheless, we caution that implicit coupling, because it leaves flexible learning space for DL with fewer physical constraints, requires more elaborate design of the model structure and inputs, as well as more thorough verification of the results to avoid learning other misleading relationships.

#### 4.2. The Physics-Informed DL Model has the Potential to Become the State-Of-Art Model in Runoff Simulation in Alpine Catchments

In this paper, the physics-informed DL model was developed in the slow change period. The results show that the model can accurately simulate multi-timescale streamflow in three subbasins with different distributions of frozen ground under the backdrop of drastic climate change and even achieve better results than the physics-based distributed model. Especially in the basin above the JIM station, where permafrost degradation is more significant, the GBEHM model completely fails to simulate both multi-timescale streamflow and the  $Q_{\max}/Q_{\min}$  ratio. In comparison, the physics-informed DL model performs well on both, as illustrated in Section 3.3.1. In order

**Table 4**  
The Relative Error (%) Between Simulated and Observed Runoff Change

| Station | Model       | Relative change of annual flow | Trend of annual flow | Relative change of the $Q_{\max}/Q_{\min}$ ratio | Trend of the $Q_{\max}/Q_{\min}$ ratio |
|---------|-------------|--------------------------------|----------------------|--------------------------------------------------|----------------------------------------|
| TNH     | EXP-HYDRO   | 78.7                           | 115.8                | /                                                | /                                      |
|         | GBEHM       | <b>5.4</b>                     | 14.0                 | /                                                | /                                      |
|         | Standard DL | 56.8                           | 86.0                 | /                                                | /                                      |
|         | Hybrid-Z    | 16.5                           | <b>12.3</b>          | /                                                | /                                      |
| MAQ     | EXP-HYDRO   | 71.0                           | 96.6                 | /                                                | /                                      |
|         | GBEHM       | 4.7                            | 27.1                 | /                                                | /                                      |
|         | Standard DL | 50.2                           | 69.5                 | /                                                | /                                      |
|         | Hybrid-Z    | <b>0.6</b>                     | <b>23.7</b>          | /                                                | /                                      |
| JIM     | EXP-HYDRO   | 83.6                           | 182.4                | 84.6                                             | 140.0                                  |
|         | GBEHM       | 33.6                           | 252.9                | 246.2                                            | 220.0                                  |
|         | Standard DL | 84.5                           | 182.4                | <b>30.8</b>                                      | 60.0                                   |
|         | Hybrid-Z    | <b>24.5</b>                    | <b>70.6</b>          | 69.2                                             | <b>20.0</b>                            |

Note. Best performers under each metric are highlighted.

to provide a more precise comparison between the physics-informed DL model and more physically-based models, literature that reported the runoff simulation results of some widely employed physically-based models at the JIM station are collected in Table 7, along with recalculated metrics for the physics-informed DL model based on the original calibration period of each referenced model without retraining. Among these models, almost all of them consider the snow cover and glacier processes, and the SWAT and VIC models further take into account the soil freeze-thaw process. As seen, the physics-informed DL model can basically achieve comparable or better performance compared to these physically-based models.

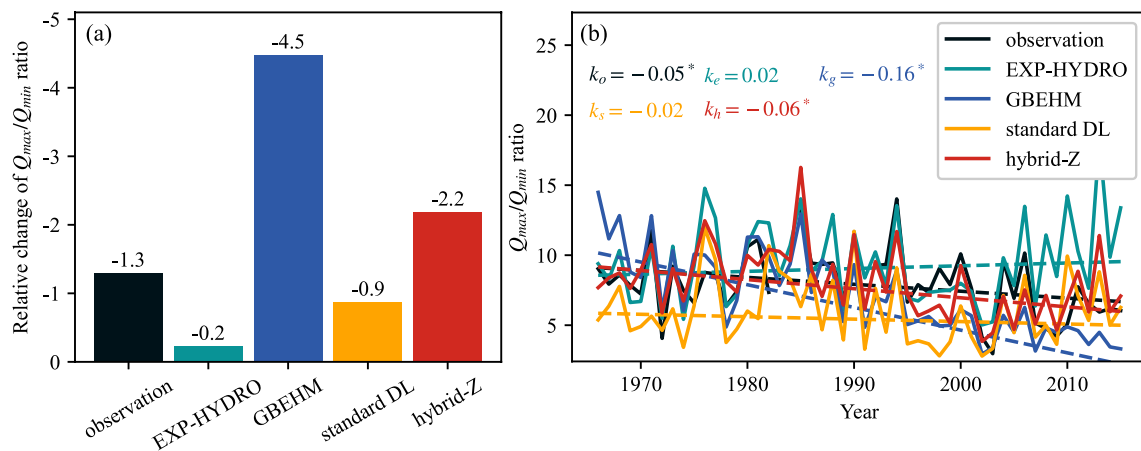
The unsatisfactory performance of the physically-based models reflects the challenges of these models in the alpine region's hydrological simulation. First of all, some process conceptualization may not be appropriate due to the insufficient understanding of alpine processes. In the case of solving the heat transfer equation, for example, the majority of models with freeze-thaw simulation typically only take into account the vertical direction and ignore the horizontal direction. Nevertheless, it has been demonstrated that lateral heat transfer can be significant for steep mountainous terrain and discontinuous permafrost (Walvoord & Kurylyk, 2016). When determining the boundary conditions, the zero-flux at the lower boundary is widely used in the models (Qin et al., 2017). However, the zero flux assumption may result in a lot of uncertainty (T. Gao, Qin, et al., 2016) since the geothermal gradients in the Qinghai-Tibet Plateau are spatially heterogeneous (Wu et al., 2010). Another major difficulty is model parameterization. Surface meteorology, borehole observations, and soil dataset are widely required for model inputs and parameterization. When the field sites are limited, some coarse-resolution, accuracy-constrained data are used for parameterization, but this may introduce great uncertainties. For instance, Qin et al. (2017) attributed the failure of the GBEHM model at the JIM station partly to the inability of the HWSO soil dataset (FAO & ISRIC, 2012) to capture the inhomogeneity of soil types well.

The comparison with other physically-based models further demonstrates the superiority of the physics-informed DL in simulating complex hydrological processes and its potential to become the state-of-art model for runoff simulation in alpine catchments. Despite this, we would like to remind that the focus of this study is on more precise streamflow prediction. In order to achieve this goal, we developed a module after the P-RNN layer that is intended to correct physical model biases, despite the potential disruption of water balance (more details could be found in Text S4 and Table S1 in Supporting Information S1). There may be potential issues when this model is utilized to analyze other hydrological variables, such as evapotranspiration and soil moisture. However, this is beyond the scope of this paper (i.e., streamflow prediction), and will be addressed in future studies. What is more significant is that beyond the model itself, the model's implicit coupling design for soil freezing and thawing, as well as its stability under climate change, may have broader implications for other catchments.

## 5. Conclusions

In this study, we took the SRYR as a case to develop a physics-informed DL model for runoff simulation in alpine catchments, which considers dominant hydrological processes, including snow accumulation and melting, soil freeze-thaw, and other fundamental hydrological processes. The two key conclusions we reached are as follows.

First, when incorporated with the physics of dominant hydrological processes, DL can simulate multi-timescale runoff and the impacts of the soil freeze-thaw



**Figure 11.** The responses of the  $Q_{\max}/Q_{\min}$  ratio to climate change at the JIM station. (a) The relative change between the rapid change period and the slow change period. (b) The trend during 1965–2015.

process on runoff very well under rapid climate change with  $1^{\circ}\text{C}$  warming. In the basins dominated by seasonally frozen soil degradation above the MAQ and TNH stations, the NSEs of the physics-informed DL model at the daily, monthly, and annual streamflow simulation can reach almost 0.9, among which the simulation of annual streamflow is better than the hybrid-J model without the representation of soil freeze-thaw. Moreover, in the basin above the JIM station, where permafrost degradation is most significant, the physics-informed DL model still outperforms the hybrid-J model not only in simulating multi-timescale streamflow, with the daily NSE of 0.77 and the monthly and annual NSE above 0.8, but also in simulating the impacts of soil freeze-thaw on runoff, with a superior reproduction on the  $Q_{\max}/Q_{\min}$  ratio and DBR event.

Second, the physics-informed DL model performs better than the lumped model, physically-based model, and standard DL model in simulating multi-timescale streamflow and the impacts of soil freeze-thaw on runoff. Among the four models, the EXP-HYDRO and standard DL models can hardly capture inter-annual hydrologic responses and thus fail at the annual streamflow simulation, and the GBEHM is not competent for runoff simulation at the JIM station. In contrast, the physics-informed DL model exhibits astounding performance on runoff simulation across all scales and stations. The physics-informed DL model also performs relatively well in simulating the  $Q_{\max}/Q_{\min}$  ratio, while other models get negative NSE. Moreover, for the simulation of the responses of annual streamflow and the  $Q_{\max}/Q_{\min}$  ratio to climate change, the physics-informed DL model comes closest to the observation. The model achieves the highest RS of 9.89 in terms of the model composite score, demonstrating its ability to best simulate runoff response under climate change. The findings suggest that incorporating dominant hydrological processes may effectively enhance the generalization ability of local DL models under climate change. This has significant implications for catchments with unique and complex hydrological processes and limited data, such as alpine regions, especially until the applicability of DL models trained on large datasets, such as CAMELS, can be verified in alpine regions.

**Table 5**  
The Rank Score for the Simulations of Multi-Timescale Streamflow, the  $Q_{\max}/Q_{\min}$  Ratio, As Well As Their Relative Change and Trend to Climate Change

| Model | EXP-HYDRO | GBEHM | Standard DL | Hybrid-Z |
|-------|-----------|-------|-------------|----------|
| RS    | 2.65      | 4.16  | 6.03        | 9.89     |

This study successfully extended the physics-informed DL to runoff simulation in the SRYR and demonstrated its ability to simulate complex hydrological processes in alpine catchments with rapid climate change, which suggests that it is a promising tool to project runoff under future climate in the alpine catchments. It should be noted that we did not consider glacial processes in the development of the physics-informed DL model because the impact of glaciers on runoff in the SRYR is negligible, but we prospect that incorporating dominant glacial processes to DL may be necessary for other alpine catchments with significant glacier melting.

**Table 6**

*The Relative Loss (%) of the Models' NSE in Multi-Timescale Streamflow Simulation on the Test Set Versus the Validation Set*

| Model       | TNH   |         |        | MAQ   |         |        | JIM   |         |        |
|-------------|-------|---------|--------|-------|---------|--------|-------|---------|--------|
|             | Daily | Monthly | Annual | Daily | Monthly | Annual | Daily | Monthly | Annual |
| Standard DL | 0.4   | 1.0     | 12.2   | 7.5   | 8.0     | 12.4   | 16.1  | 13.3    | 30.6   |
| Hybrid-J    | 2.6   | 2.4     | 6.2    | 4.1   | 4.1     | 8.7    | 13.1  | 8.9     | 17.3   |
| Hybrid-Z    | 1.3   | 1.1     | 3.2    | 2.7   | 1.9     | −3.9   | 11.4  | 6.9     | 1.6    |

**Table 7**

*Comparison of the Physics-Informed DL Model With Other Physically-Based Models at the JIM Station*

| Physics-informed DL model |             |        | Referenced physically-based model |          |       |         |                   |
|---------------------------|-------------|--------|-----------------------------------|----------|-------|---------|-------------------|
| NSE                       | <i>r</i>    | PBIAS  | NSE                               | <i>r</i> | PBIAS | Scale   | Model             |
| 0.70                      | 0.86        | −8.7%  | 0.71                              | 0.89     | 23.0% | Daily   | HEQM <sup>a</sup> |
| <b>0.79</b>               | <b>0.90</b> | −7.2%  | 0.39                              | 0.75     | /     | Daily   | SWAT <sup>b</sup> |
| <b>0.74</b>               | 0.89        | −10.6% | 0.68                              | 0.89     | 15.4% | Daily   | SWAT <sup>c</sup> |
| <b>0.90</b>               | 0.96        | −4.1%  | 0.75                              | /        | /     | Monthly | VIC <sup>d</sup>  |
| <b>0.90</b>               | <b>0.97</b> | −10.3% | 0.77                              | 0.88     | −3.3% | Monthly | SWAT <sup>e</sup> |
| <b>0.84</b>               | <b>0.92</b> | −4.7%  | 0.72                              | 0.85     | −1.0% | Monthly | SPHY <sup>f</sup> |

*Note.* The comparative period is unified as the original calibration period of each model in the literature. The metric by which the physics-informed DL model outperforms the reference model is highlighted.

<sup>a</sup>The data source: W. Wang et al. (2021). <sup>b</sup>The data source: Gu et al. (2022). <sup>c</sup>The data source: K. Li, Liu, et al. (2021). <sup>d</sup>The data source: S. Li, Liu, et al. (2021). <sup>e</sup>The data source: Zeng et al. (2022). <sup>f</sup>The data source: T. Zhang et al. (2022).

## Data Availability Statement

The meteorological data used in this paper can be obtained from <http://data.cma.cn/data/detail/data-Code/A.0012.0001.html>. The SRTM DEM data (Reuter et al., 2007) can be downloaded from <https://srtm.csi.cgiar.org/srtmdata/>. The source code and models' simulated streamflow in this study are available in the Zenodo repository via <https://zenodo.org/record/8041812> (Zhong et al., 2023).

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